Pandas



Data Aggregating

BEGINNER COURSE — 2025

PYTHON FOR DATA SCIENCE

Smiaansh



PANDAS

DATA AGGREGATING

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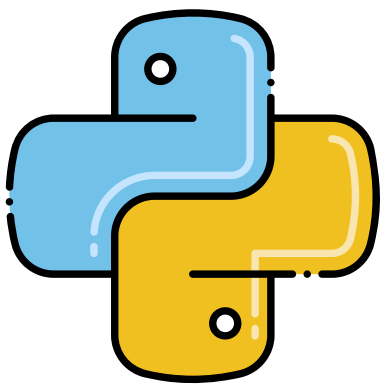


Aggregating

- Aggregation = Reducing massive rows into meaningful metrics like:
  - Totals
  - Averages
  - Counts
  - Maximums/Minimums
- It helps uncover patterns that guide real-world decisions.

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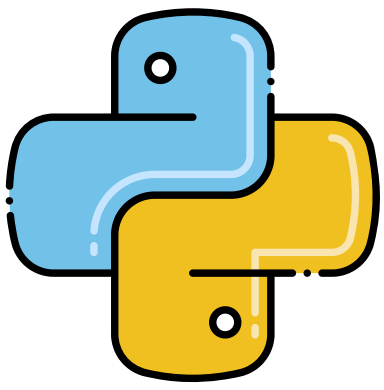
Real Life Examples

- 🛍️ Avg cart value per customer (E-commerce)
- 🚚 Total distance per driver (Logistics)
- 👨‍⚕️ Avg wait time per department (Healthcare)
- 🎧 Top tracks by user (Streaming apps)
- 🧾 Revenue per category (Retail analytics)

✅ Every dashboard you admire uses aggregation!

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Aggregating Data

```python
"""
Example: Pandas Aggregation Data
"""

import pandas as pd

# Read the Raw CSV Data
df = pd.read_csv("path_to_csv")

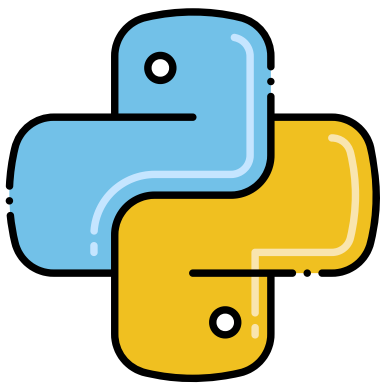
df['sales'].sum()                # Total
df['rating'].mean()              # Average
df['orders'].count()             # Count
df.groupby(
    'region'
)['profit'].agg(['sum', 'mean'])

```

✅ Use .agg() for custom multiple aggregations

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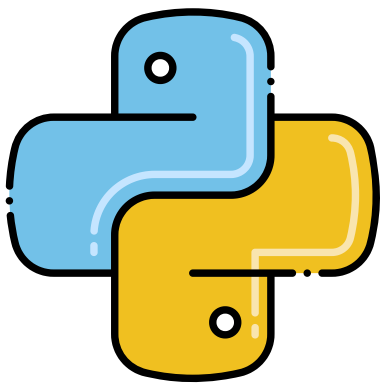
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Why is this powerful?

- Convert massive data into more clarified and summarized data that is easy to understand
- Sumarized data can be visualized easily
- Every dashboard with charts or some kind of visualization uses aggregation behind the scene.

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📌 Your Data Deserves to Be Heard

- It’s not about collecting more data — it’s about learning to ask better questions.

🔍 Aggregation is how you listen to what your data is trying to say.

- ➡ Visualization + Storytelling

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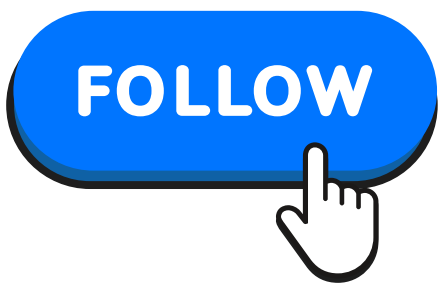


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Fun Fact

- Pandas aggregation works on more than numbers!
- You can aggregate strings, dates, and even text:

```
df.groupby('author')[
    'comments'
].agg(lambda x: ' | '.join(x))
```



me to know more about pandas!

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